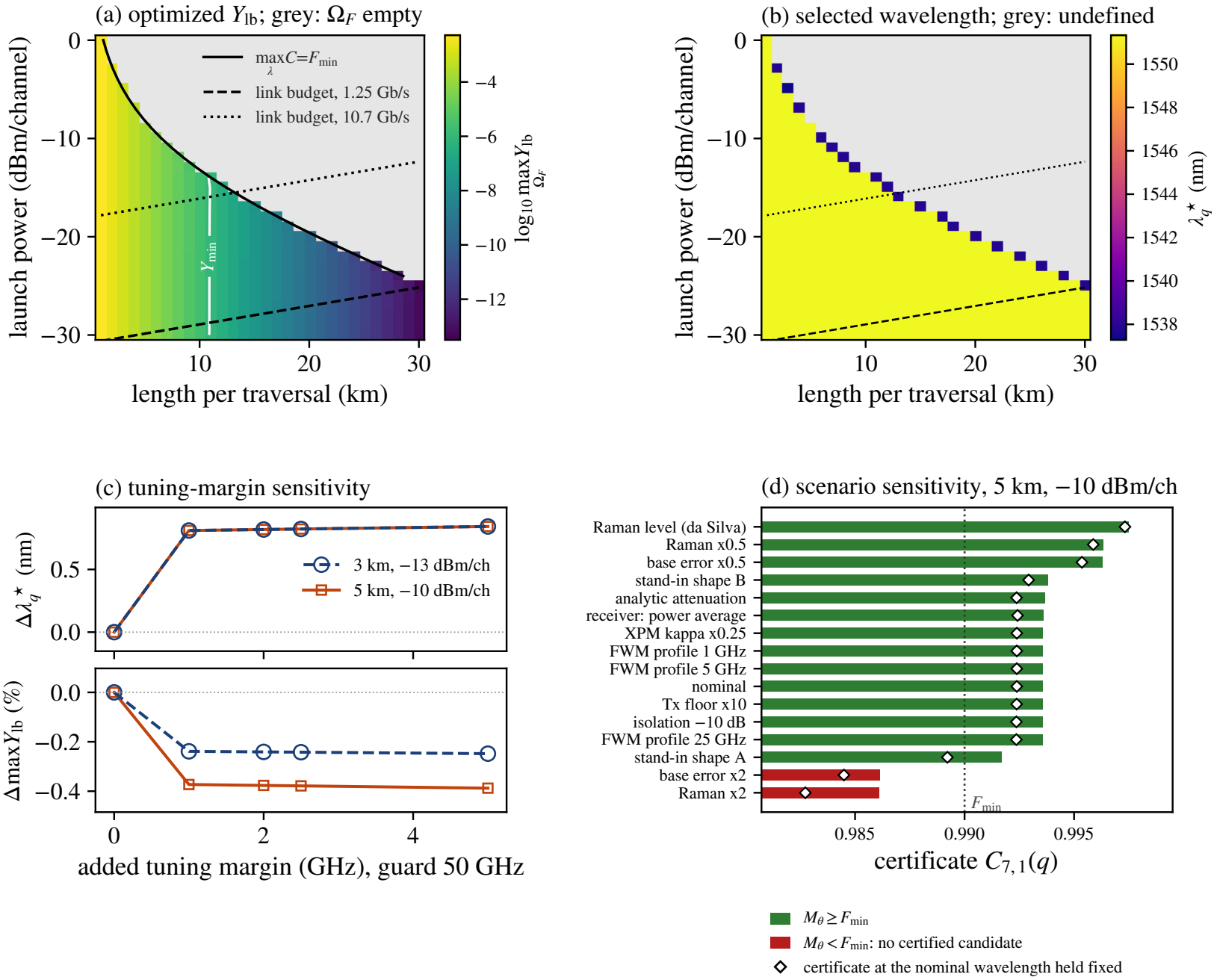
(a) optimized Y_{lb}; grey: Ω_F empty
launch power (dBm/channel)
length per traversal (km)
$\max_{\lambda} C = F_{\min}$
link budget, 1.25 Gb/s
link budget, 10.7 Gb/s
$Y_{\min}$
$\log_{10} \max_{\Omega_F} Y_{\mathrm{lb}}$
(b) selected wavelength; grey: undefined
launch power (dBm/channel)
length per traversal (km)
$\lambda_q^\star$ (nm)
(c) tuning-margin sensitivity
$\Delta\lambda_q^\star$ (nm)
3 km, −13 dBm/ch
5 km, −10 dBm/ch
$\Delta \max Y_{\mathrm{lb}}$ (%)
added tuning margin (GHz), guard 50 GHz
(d) scenario sensitivity, 5 km, −10 dBm/ch
Raman level (da Silva)
Raman x0.5
base error x0.5
stand-in shape B
analytic attenuation
receiver: power average
XPM kappa x0.25
FWM profile 1 GHz
FWM profile 5 GHz
nominal
Tx floor x10
isolation −10 dB
FWM profile 25 GHz
stand-in shape A
base error x2
Raman x2
$F_{\min}$
certificate $C_{7,1}(q)$
$M_\theta \geq F_{\min}$
$M_\theta < F_{\min}$: no certified candidate
certificate at the nominal wavelength held fixed